

Safid

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Spring-2025

Course Title: Database Management System

Course Code: CSE 315

Marks: 50

Time: 3(three) hours

(Answer all questions sequentially)

Marks

1. a) Define deadlock in DBMS and list the conditions for it to occur. [05]
 b) Perform left outer join, right outer join, full outer join and inner join on the Patient(table-1) and Patient_due (table-2). [05]

Table 1: Patient

Table 2: Patient_due

Patient_id	P_Name
P101	Siam
P102	Rana
P103	Tuhin

admission_date	Due	Patient_id
2025-01-15	15000	P101
2025-03-10	10000	P102
2025-05-25	5000	P105

2. a) Consider the following relational database, [03]

Project (ProjID, ProjName, DeptID)

WorksOn (EmpID, ProjID, Hours)

Write the Relational Algebra (RA) for each of the following statement:

- Find the EmpIDs of employees who work on the project named 'Apollo'.
- List the ProjIDs of projects that have employees working more than 40 hours
- Find the EmpIDs of employees who work on at least one project handled by Department 10.

- b) Consider the following relational schema: [07]

Employee = (EmpID, EmpName, Department, Project, Address, gender)

Employee_position = (EmpID, EmployeePosition, DOJ, Salary)

Develop the MySQL query to answer the following question:

- Show Department wise Employee list from Employee table
- To display the EmpName and Department whose salary is greater than 10000.

- iii. Find duplicate employee records from the employee table.
- iv. Increase the 5% salary of all employees whose salary is less than 10000.
- v. Find the second highest salary.
- vi. Fetch the Name of the employee with the same salary.
- vii. Find Top 10% row from Employee table based on salary.

3. a) A library management system uses the following unnormalized table (table-3) to track book loans. Identify three anomalies (insertion, deletion, update) in the Book_Loans table and explain each with an example. [05]

Table 3: Book_Loans

Loan id	Book title	author	Student id	Student name	Loan date	Return date	Dept
L001	DBMS	Smith	S1001	John	2023-09-01	2023-09-15	CSE
L002	OOP	Brown	S1002	Alice	2023-09-05	2023-09-20	IT
L003	DBMS	Smith	S1003	Bob	2023-09-10	NULL	CSE
L004	OOP	Davis	S1001	John	2023-09-12	NULL	CSE

- b) Design a fully normalized database schema for the library management system. [05]

4. a) A developer argues that normalization slows down queries. **Evaluate** this claim by discussing: [05]

1. When normalization improves performance.
2. When denormalization might be preferred.

- b) Draw a Schema diagram of an Admission System at UITS including cardinality. [05]

5. a) Define the Wait-Die and Wound-Wait schemes in one sentence each. Explain the key difference between them? [03]

- b) Draw an ER diagram of a Registration System at UITS. The diagram should contain strong entity, weak entity, key attribute, multi-valued attribute, composite attribute, derived attribute. [04]

- c) Describe how starvation might occur in a DBMS that uses a first-come-first-serve (FCFS) scheduling policy for resource allocation, and discuss its impact on transaction fairness. [03]